Topic

Predicting credit card customer churn using machine learning models.

Business Problem

The banking industry is highly competitive, and acquiring new credit card customers is both difficult and costly. At the same time, losing existing customers leads to a direct loss of revenue. Once a customer switches to another provider, it is unlikely they will return.

Customer retention is critical because it helps maintain a stable customer base and ensures recurring revenue. This project aims to identify customers who are likely to stop using their credit cards so that the bank can take proactive actions to retain them.

Research Questions

How accurately can machine learning models predict customer churn?

What factors contribute the most to customer churn?

Are there identifiable behavioral patterns among customers who leave?

How can the results be used to improve customer retention strategies?

Dataset

The dataset used in this project is the *BankChurners* dataset obtained from Kaggle.

The dataset contains customer-level information related to credit card usage and churn behavior. It includes demographic variables (such as age, gender, and income), account details, and transaction activity metrics.

The dataset used for this analysis contains 10,128 rows and 23 columns. Key variables include:

Customer demographics (age, gender, education, marital status, income)

Account information (credit limit, months on book, card category)

Transaction behavior (total transactions, transaction amount)

Activity measures (inactive months, contact frequency)

Target variable: *Attrition_Flag* (indicating whether a customer has churned)

Methods

- Data cleaning and preprocessing
- Exploratory Data Analysis (EDA)
- Classification models:
    - Logistic Regression
    - Random Forest
    - XGBoost
- Model evaluation using accuracy and ROC-AUC
- Feature importance analysis to identify key drivers of churn

(Note: Methods may be adjusted as the analysis progresses.)

Ethical Considerations

- Potential bias in customer segmentation models
- Ensuring fair treatment across demographic groups
- Avoiding discrimination based on sensitive attributes such as income or gender
- Responsible use of predictions to support customers rather than penalize them

Challenges / Issues

- Potential class imbalance in the dataset (to be evaluated)
- Selecting the most relevant features for modeling
- Risk of omitting important variables
- Ensuring model interpretability for business use

References

Kaggle dataset: BankChurners

Ngai, E. W. T., Xiu, L., & Chau, D. C. K. (2009). *Application of data mining techniques in customer relationship management: A literature review.*

Verbeke, W., Dejaeger, K., Martens, D., Hur, J., & Baesens, B. (2012). *New insights into churn prediction using data mining techniques.*